A Triplet Semisupervised Deep Network for Fusion Classification of Hyperspectral and LiDAR Data

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***Abstract*— Data fusion of hyperspectral and light detection and ranging (LiDAR) is conducive to obtain more comprehensive surface information and thereby achieve better classification result in Earth monitoring systems. However, lack of labeled samples usually limits the performance of supervised classifiers, and the heterogeneity of multisource data also brings great challenges to data fusion. Aiming to address these issues, we propose a triplet semisupervised deep network (TSDN) for fusion classification of hyperspectral and LiDAR. Specifically, we utilize three basic pathways to extract deep learning features: 1-D convolutional neural network (CNN) for spectral features in hyperspectral, 2-D CNN for spatial features in hyperspectral, and Cascade Net for elevation features in LiDAR data. Furthermore, a novel label calibration module (LCM) is proposed to generate effective pseudo labels with high confidence based on the superpixel segmentation by comparing the multiview classification results for assisting semisupervised model training. In addition, we design a novel 3D-Cross Attention Block to enhance the complementary spatial features of multisource data. Experiments on three public HSI–LiDAR benchmarks, Houston, Trento, and MUUFL Gulfport, have demonstrated the effectiveness and superiority of our proposed method.**

***Index Terms*— Classification, data fusion, deep learning, hyperspectral image, semisupervised.**

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I. Introduction

HYPERSPECTRAL imagery (HSI) contains a wealth of spectral information that facilitates a more subtle identification of remote sensing objects [1], [2], [3], [4]. In recent years, plenty of traditional machine learning-based classification methods have been proposed to extract spectral information from HSI to distinguish different classes of ground surface objects [5]. For example, by measuring the spectral differences between the test and the training samples, K-nearest neighbor (KNN) determines the category. Support vector machine (SVM) is based on nonlinear mapping to high-dimensional space through utilizing kernel function and obtains stable classification performance [6]. In comparison to SVM, extreme learning machine (ELM) performs nearly as well as it but with a lower computation cost [7]. With the development of deep learning, convolutional neural networks (CNNs) have been proved more effective than traditional methods [8]. The CNN-PPF proposed by Li et al. [9], [10] achieves superior accuracy due to its excellent spectral feature extraction capability. Despite this, all aforementioned methods ignore the importance of spatial context information in hyperspectral classification. Thus, the 3-D CNN by Chen et al. [11] directly extracted spatial–spectral joint features of original hyperspectral cube with a relatively simple network architecture. Xi et al. [12] built ensemble deep kernel extreme learning machine (KELM) to investigate the deep-level characteristics in the superpixel pattern and attentional neighboring superpixel pattern. The spectral-spatial residual network (SSRN) by Zhong et al. [13] introduces the idea of residual network based on 3-D convolution, which greatly improves the classification performance. Wang et al. [14] extracted the multiscale contextual information of HSI by an end-to-end adaptive spectral–spatial network, which achieves competitive results. Furthermore, the adjacency matrix can be established by using the association information between samples and input into the graph convolution network (GCN), so as to realize the discrimination of ground object attributes from the perspective of graph theory [15]. Xi et al. [16] used a cross-scale graph prototypical network (X-GPN) to achieve semisupervised high-quality HSI classification. Li et al. [17] developed the SGML to combine multiscale G-Cov and C-Cov, which achieves a better classification result. To further enhance the performance with few labeled samples, Xi et al. [18] proposed a few-shot learning framework with class-covariance metric (CMFSL) for HSI classification, which can outperform the

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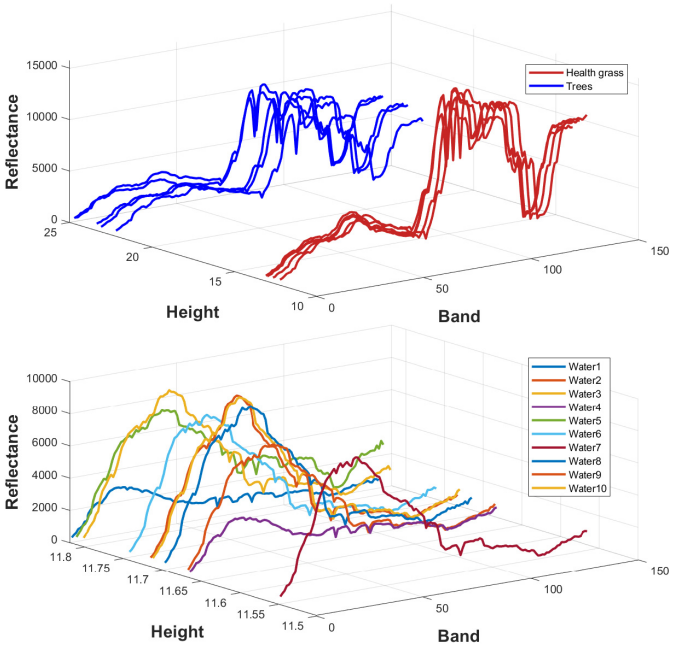


Fig. 1. Importance of height information to two types of spectral aberration in the Houston2013 dataset.

state-of-the-art methods with few-shot annotated samples. Considering the limitation of CNNs' inherent network backbone, Hong et al. [19] proposed the SpectralFormer to mine and represent the sequence attributes of spectral signatures and got a significant improvement in comparison with other HSI classification methods.

As a result of the phenomenon of same object with different spectral (SODS) and the phenomenon of same spectrum with different objects (SSDO), the classification performance achieved solely by HSI is greatly perturbed. In response to researchers' demands for classification accuracy increase, remote sensing technologies allowing more types of information to be obtained about the surface, which promotes the development of multisensor fusion [20]. Light detection and ranging (LiDAR) carries surface elevation information with little interference from atmospheric conditions. Thus, it is justified to classify HSI and LiDAR as one, as their complementary information greatly enhances each other [21]. As shown in Fig. 1, SODS and SSDO can be effectively alleviated in the HSI–LiDAR fusion classification task [21], [22], [23], [24]. For instance, based on the height information provided by LiDAR, the classification algorithms can accurately distinguish SSDO, such as health grass and trees. Meanwhile, the reference of height information also alleviates SODS, such as water.

Fusion classification of HSI and LiDAR is full of challenges by the differences in data characteristics acquired by different sensors [25]. Image fusion is divided into three levels, pixel-level fusion, feature-level fusion, and decision-level fusion [26], usually corresponding to the first, middle, and last stages of dataflow, thus guiding the design of HSI–LiDAR fusion algorithm. For instance, Xu et al. [27] proposed the two-branch CNN, which takes the lead in using the deep learning method to achieve feature-level fusion of HSI and LiDAR. In the two-branch CNN, the dual-tunnel CNN branch extracts spatial features and spectral information carried by HSI, the cascade CNN branch extracts LiDAR information, and the method of different branches extracting multisource data features lays a foundation for subsequent algorithms. Compared with traditional methods, the two-branch CNN shows obvious advantages. However, the two-branch CNN only realizes fusion by concatenating features and does not give full play to the potential of multisource data fusion and classification. For better deep learning features fusion, Zhang et al. [28] designed the PToP CNN with an unsupervised feature extractor to seek multiscale features expression with a three-tower structure, but it still relies on label information in the decision stage and less consideration on introducing attention modules or residual blocks also limits the performance. Aiming to making full use of multisource data, Mohla et al. [29] built the FusAtNet to introduce the mutual attention mechanism between hyperspectral and LiDAR to realize feature enhancement and improve the effect of fusion classification, but the complex network structure greatly increases the computation cost of the FusAtNet. The above fusion classification algorithms focused on single-level fusion, but the most advanced research shows the superiority of multilevel fusion strategies [30], [31]. From the image processing perspective, Zhao et al. [32] designed the HRWN; it applies the Gram–Schmidt spectral sharpening method to perform pixel-level fusion in the input stage, and then, the random walk algorithm is used for performing decision-level fusion in the output stage, thus achieving a high-precision classification result, but the simple network of the HRWN leaves room for improvement on deep learning model. The A^3CLNN is designed for a dual-channel spatial, spectral, and multiscale attention convolutional long short-term memory neural network by Li et al. [33], and the A^3CLNN not only contains modules of spatial attention, spectral attention, and multiscale residual attention but also introduces a three-level fusion strategy, so it has strong feature representation capability and well classification performance, but relatively speaking, it also increases the risk of model overfitting.

In summary, fully supervised deep learning-based algorithms are usually not a feasible solution to achieve superior performance without a vast amount of training data [34]. In fact, labeling HSI data is expensive and time-consuming, thus limiting the application of these fully supervised algorithms [35]. In the field of natural image processing such as semantic segmentation, the semisupervised learning method effectively solves the problem of lacking labeled samples [36]. For example, recently, self-training has been widely used in semisupervised semantic segmentation tasks, which first obtains a pretrain model on labeled data and then makes predictions based on the model for unlabeled data [37]. The segmentation result of unlabeled data by the pretrain model is called pseudo labels. Then pseudo labels are added to retrain the segmentation model, therefore reaching satisfactory segmentation results [38]. For HSI and LiDAR fusion classification tasks, manual pixel-level annotation is more difficult. Thus, it is valuable and urgent to explore the available unlabeled samples to help learn classification models. Meanwhile, how to improve the complementarity of spatial information of multisource data is also the key to improving the performance

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of the classification algorithm. Currently, popular attention mechanisms are generally designed for single sensor images, and with little consideration given to how to balance spatial information from HSI and LiDAR, this impairs the advantages of the attention mechanism in the field of HSI–LiDAR classification. Target on the above problems, we proposed a triplet semisupervised deep network (TSDN), not only introducing a novel pseudo label acquisition method for semisupervised learning to expand available train samples but also designing a unique attention module to enhance the model's representation of deep information. As far as we know, our method is the first semisupervised classification deep learning-based method in the HSI–LiDAR fusion classification task.

In summary, the main contributions can be summarized as follows.

1) We propose a triplet branch fusion classification network named TSDN, which consists of 1-D CNN for spectral features extraction in HSI, 2-D CNN for spatial features extraction in hyperspectral, and Cascade Net for elevation features extraction in LiDAR data. In particular, TSDN combines feature-level fusion and decision-level fusion strategies to achieve efficient multisource data fusion and classification.
2) To overcome the heterogeneity and make full use of the spatial complementarity of HSI and LiDAR in fusion classification, a 3-D Cross Attention Block is designed for the feature-level fusion of HSI and LiDAR. We consider that the joint space information has important differences not only in spatial dimension but also in channel dimension; therefore, our proposed 3D-Cross Attention Block is uniquely comprising two branches to extract attention masks in spatial and channel dimensions. Then, we implement the simultaneous allocation of attention to form 3-D attention. 3D-Cross Attention Block effectively improves the performance of the classifier.
3) To increase the available labeled samples of HSI–LiDAR fusion classification task, we put forward the label calibration module (LCM), which introduces GCN's prediction result from a graph perspective to generate multiview output to discriminate the consistency of each sample. Then, LCM produces pseudo labels for each sample according to multiview output and superpixel map. In the end, the original real labels and pseudo labels are mixed to complete a semisupervised training process.

The leftover parts of this article are arranged in the following manner. Section II presents a brief description of the related work. Section III details the method we propose. Section IV carries out plenty of experiments and analyses of the classification results. Toward the end of this article, the conclusion is drawn in Section V. The leftover parts of this article are arranged in the following manner. Section II presents a brief description of the related work. Section III details the method we propose. Section IV carries out plenty of experiments and analyses of the classification results. Toward the end of this article, the conclusion is drawn in Section V.

II. Related Work

A. Semisupervised Learning

It is a key field in machine learning, which shows great advantages in the condition of being short of labeled samples because semisupervised learning not only extracts the information of labeled samples but also makes full use of the features of unlabeled samples [39], [40]. In order to make full use of unlabeled samples, semisupervised learning requires the assumption that data distribution information is associated with category labels. "Clustering assumption" and "Manifold assumption" are the most common assumptions in semisupervised learning [41], [42]. Clustering assumption supposes that the data have a cluster structure and the samples of the same cluster belong to the same category, and manifold assumption supposes that the data are distributed over a manifold structure and adjacent samples have similar outputs. According to the above assumptions, a semisupervised learning algorithm can still obtain the information about the data distribution of unlabeled samples, which is very beneficial to the establishment of classifier model. Therefore, on the one hand, compared with supervised learning methods, semisupervised learning has better generalization; on the other hand, semisupervised learning has obvious advantages over unsupervised learning in classification accuracy.

B. Attention Mechanism

Bahdanau et al. [43] were the first to combine the attention mechanism with deep learning and made great progress in the field of natural language processing (NLP); subsequently, attention mechanism is widely used in the field of computer vision (CV). The main principle of attention mechanism is to allocate the features extracted by the neural network according to their importance by learning a weight map [44]. Considering the different importance of feature maps in channel dimension, Hu et al. [45] first proposed the squeeze-and-excitation (SE) module to realize the channel attention allocation. Taking X as input and Y as output, denote $\otimes$ as elementwise multiply, and SENet can be formulated as

$$A_{\mathrm{SE}} = \sigma\left(W_2\delta\left(W_1\,\mathrm{GAP}(X)\right)\right)$$
$$Y = A_{\mathrm{SE}} \otimes X.$$

In fact, feature maps also have different importance in terms of spatial dimensions, Woo et al. [46] proposed a convolutional block attention module (CBAM), which is the first attention module that merges spatial and channel attention. CBAM has two sequential submodules: channel attention submodule and spatial attention submodule. CBAM further improves the representation ability of neural network. Taking X as input and Y as output, CBAM can also be formulated as

$$A_{\mathrm{chan}} = \sigma\left(W_2\delta\left(W_1\,\mathrm{GAP}(X)\right) + \left(W_2\delta\left(W_1\,\mathrm{GMP}(X)\right)\right)\right)$$
$$X' = A_{\mathrm{chan}} \otimes X$$
$$A_{\mathrm{spat}} = \sigma\left(\mathrm{Conv}\left(\left[\mathrm{GAP}\left(X'\right);\mathrm{GMP}\left(X'\right)\right]\right)\right)$$
$$Y = A_{\mathrm{spat}} \otimes X'.$$

In general, in the above attention module, the output feature Y makes the useful information in the input feature X

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more obvious, and the distribution of similar substances in the feature space can be more concentrated, which increases the prediction reliability of the classifier. Inspired by the attention mechanism, we design a simple but effective attention module, which follows the method of merged attention, extracts attention in channel dimension and spatial dimension, and combines it into 3-D attention.

C. Superpixel Segmentation

Superpixel segmentation divides the image into multiple subregions by clustering adjacent spatial pixels with similar spectral features [47]. Superpixels contain many benefits for classification tasks [48], such as homogeneity, boundary adherence, regularity, and connected partition. Current superpixel segmentation algorithms can be categorized into two groups: graph-based and gradient-descent-based. On the one hand, Achanta et al. [49] proposed the simple linear iterative clustering (SLIC) algorithm, which is most suitable for processing hyperspectral data among gradient-descent-based algorithms [49]. On the other hand, among the graph-based superpixel algorithms, the entropy rate superpixel (ERS) algorithm proposed by Liu et al. [50] is most widely used in the hyperspectral field.

As the most classical hyperspectral image superpixel segmentation algorithm, the SLIC algorithm is briefly described as follows: we denoted input HSI as $H \equiv \{P_1^b, P_2^b, \ldots, P_n^b\}$ with n pixels, and P_i^b denotes the value at the ith pixel for the bth spectral band and $i = 1, 2, \ldots, n$, and $b = i = 1, 2, \ldots, B$.

First, the distance metric of cluster center and pixel i is defined as

$$\mathfrak{D} = \mathfrak{D}_{\text{spec}} + \frac{w}{R}\mathfrak{D}_{\text{spat}}$$

where R is calculated by $2(n/K)^{1/2}$, w is the weighting factor between spectral and spatial features, $\mathfrak{D}_{\text{spec}}$ describes the measure of homogeneity, and $\mathfrak{D}_{\text{spat}}$ ensures compactness and regularity in superpixels.

Then, the spectral distance metric can be represented as

$$\mathfrak{D}_{\text{spec}} = \sqrt{\sum_{i=1}^{B}\left(P_i^b - P_j^b\right)^2}$$

and the spatial distance metric can also be represented as

$$\mathfrak{D}_{\text{spat}} = \sqrt{\left(r_i - r_j\right)^2 + \left(u_i - u_j\right)^2}$$

where $(r; u)$ denotes the location of pixel i in superpixel.

III. Methodology

In this section, we develop the network framework of TSDN. For multisource heterogeneous data, TSDN uses different network structures to extract features and then does feature-level fusion. LCM can obtain high-confidence pseudo labels without manual labeling of samples and help backbone networks to complete semisupervised learning. In particular, the 3D-Cross Attention Block enhances the complementary spatial features of multisource data and increases the robustness of the classifier. We will introduce the design principle of these modules in turn.

A. Network Architecture

As shown in Fig. 2, TSDN consists of CNNs with three branches of 1-D CNN, 2-D CNN, and Cascade Net as the basic feature extractor. It is worth mentioning that triplets do not refer to GCN and superpixel branches since they are only auxiliary pseudo-label generation and do not participate in the feature extraction process of the model. Among triplet networks, 1-D CNN takes spectral vectors as input to extract spectral features, 2-D CNN takes HSI patches as input to extract spatial features from his, and CascadeNet takes LiDAR patches as input to extract spatial elevation features from LiDAR. The parameter details of these networks are shown in Fig. 3. We concatenate the spatial information extracted from multisource heterogeneous data to form complementary spatial features. Then, for spectral features and spatial features, we use different attention modules to realize the enhancement of deep features, so as to better complete the classification task.

TSDN uses unique dataflows to achieve semisupervised learning. First, according to the center of train samples, we divide the data into patches, and then, we obtain a pretrain model and make predictions for the test samples; meanwhile, we train a graph convolutional network and obtain its prediction results. Different from CNNs, graph convolution is based on the adjacency matrix of labeled samples to predict test sets. Therefore, we obtain multiview prediction results. Then, the multiview prediction results are input into the LCM module, and LCM will generate some pseudo labels through specific algorithms. Finally, the real labels of the train set and the pseudo labels of the testing set are mixed together to realize the expansion of the training sample and realize the semisupervised training of the network.

The base triplet convolution neural network extracts the feature as the following formula:

$$\mathbf{F}\left(\mathbf{X}_{\text{HSI}}, \mathbf{X}_{\text{LiDAR}}\right) = f\left(\mathbf{W} \cdot \left(\mathbf{F}_{\text{spec}} \otimes A_{\text{spec}} \middle\| \mathbf{F}_{\text{spat}} \otimes A_t\right) + \mathbf{b}\right).$$

Then, the softmax layer predicts the probability distribution by

$$p_i^k = \frac{\exp\left(\theta_k \middle| \mathbf{F}\left(\mathbf{X}_{\text{HSI}}, \mathbf{X}_{\text{LiDAR}}\right)\right)}{\sum_{k=1}^{K}\exp\left(\theta_k \middle| \mathbf{F}\left(\mathbf{X}_{\text{HSI}}, \mathbf{X}_{\text{LiDAR}}\right)\right)}$$

where K is the number of classes, k indicates the kth class, i is the index of pixel, and θ_k is the kth column of the weights in the prediction layer.

B. Label Calibration Module

The proposed LCM module cleverly utilizes the different design ideas of graph convolutional network and normal CNN. Since the graph convolutional network deduces from the relationship between graph nodes, while deep neural network focuses on the pixel value of a certain pixel point and its neighbor pixel points, from the consistency and difference of the classification results of these two networks, we define two types of samples: easy samples and hard samples. Fig. 4 shows how LCM generates pseudo labels for these two types of samples.

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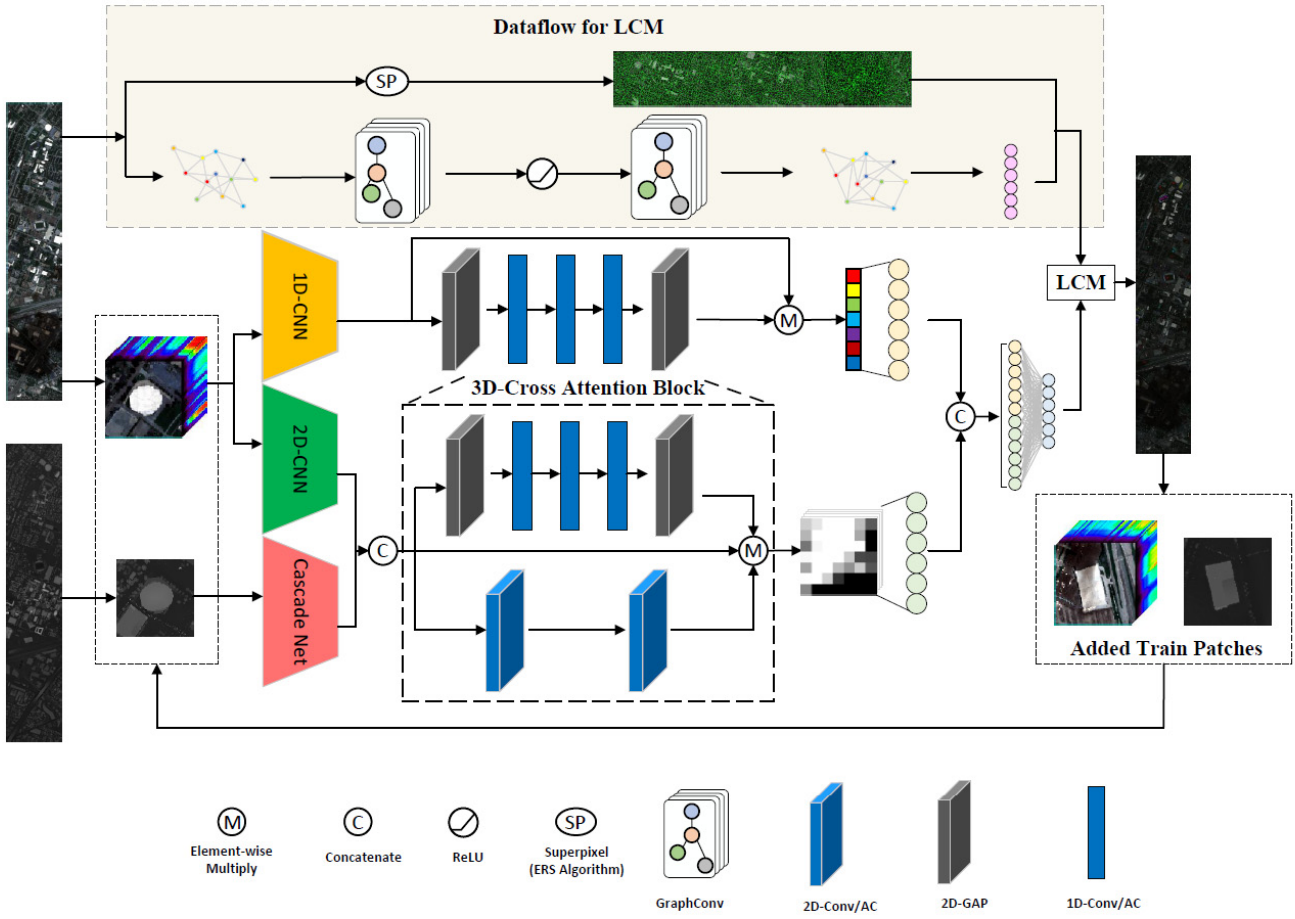


Fig. 2. Overview of the proposed TSDN. The core structure of TSDN uses a three-branch CNN as a feature extractor to extract spectral, spatial, and elevation information from multisource input and to enhance the extracted depth features by using the attention mechanism. Hyperspectral data are also used for GCN classification and superpixel generation as auxiliary information input to LCM to generate pseudo labels. Finally, the patches of network training are added with the generated pseudo labels to complete the semisupervised training process.

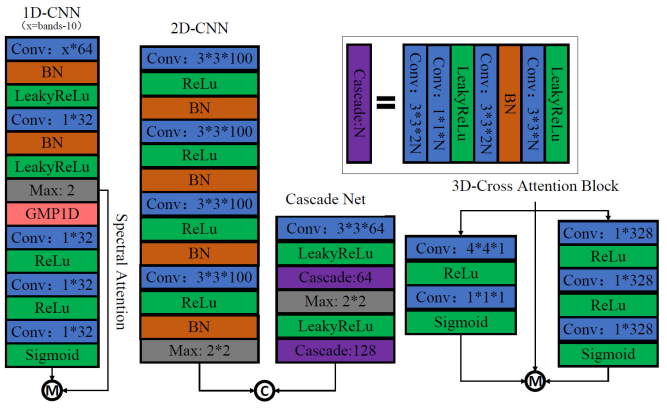


Fig. 3. Details of the feature extractor. Among them, 1-D CNN extracted spectral features of hyperspectral data, 2-D CNN extracted spatial context information of hyperspectral data, Cascade Net extracted elevation information of LiDAR data, and 3D-Cross Attention Block enhanced the fused spatial features from HSI and LiDAR.

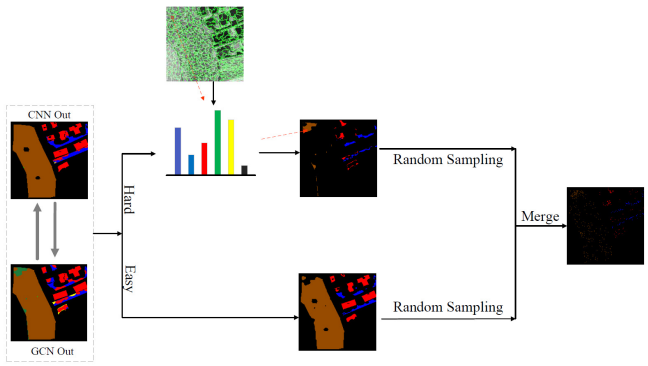


Fig. 4. By comparing the different classification results of GCN and CNN, the testing set samples were divided into hard samples and easy samples. We used targeted methods to generate pseudo labels of these samples. The pseudo labels of easy samples are the classification results of two networks; the pseudo labels of hard samples are determined by MV through the CNN output results in superpixel. Finally, a certain proportion of hard samples and easy samples were randomly selected for mixing as the output of LCM.

We select the most robust ERS superpixel segmentation algorithm from the graph-based methods as the prior information of LCM. The formula of ERS algorithm is described in Algorithm 1.

According to the input HSI data, let V represent the set of vertices v_i for each pixel in the image and E represent the set of edges e_{ij} between v_i and v_j in V; we construct graph $\mathcal{G} = (V, E)$, and ERS aims to find a graph $\mathcal{G} = (V, E')$

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Algorithm 1 Description for LCM Algorithm

Input: $Y_c = \{c_1, c_2, \ldots, c_n\}$, $Y_g = \{g_1, g_2, \ldots, g_n\}$, $S = \{s_1, s_2, \ldots, s_m\}$
Output: $P = \{p_1, p_2, \ldots, p_a\}$
for $i = \{1, 2, \ldots, n\}$ **do**
  if $c_i = g_i = k, k = 1, 2, \ldots, C$ **then**
    $p_i = k$;
  end
  else
    s_j is the superpixel of c_i;
    $V[K] = [0, 0, \ldots, 0]$;
    foreach c_x *in* s_j **do**
      $V(c_x) += 1$;
    end
    $p_i = max(V[K])$
  end
end
return $RandomSamples(p_i)$;

that divides the image into K connected subgraphs from $\mathcal{G} = (V, E)$. From the spectral distance between neighbor pixels, we calculate a weight denoted as w_{ij}. Then, the entropy rate function $\mathcal{H}(E')$ can be expressed as follows:

$$\mathcal{H}\left(E'\right) = -\sum_i \mu_i \sum_j P_{i,j}\left(E'\right) \log\left(P_{i,j}\left(E'\right)\right)$$

where $P_{i,j}$ is the transition probabilities in which the expression is given as follows:

$$P_{i,j}\left(E'\right) = \begin{cases} \dfrac{w_{i,j}}{w_i}, & \text{if } i \neq j \text{ and } e_{i,j} \in E' \\ 0, & \text{if } i \neq j \text{ and } e_{i,j} \notin E' \\ 1 - \dfrac{\sum_{j:e_{i,j}\in E'} w_{i,j}}{w_i}, & \text{if } i = j \end{cases}$$

$$w_i = \sum_{k:e_{i,k}\in E} w_{i,k}, \quad w_T = \sum_{i=1}^{|V|} w_i, \quad \mu_i = \frac{w_i}{w_T}.$$

A balancing function was also introduced as follows:

$$\begin{aligned} \mathcal{B}\left(E'\right) &= \mathcal{H}\left(Z_{E'}\right) - \mathcal{N}_{E'} \\ &= -\sum_i PZ_{E'}(i) \log\left(PZ_{E'}(i)\right) - \mathcal{N}_{E'} \end{aligned}$$

where

$$PZ_{E'}(i) = \frac{|S_i|}{|V|}, \quad i = \{1, \ldots, N_{E'}\}.$$

Finally, the objective function is defined as

$$\max_{E'} \mathcal{H}\left(E'\right) + \lambda \mathcal{B}\left(E'\right)$$

subject to $E' \subseteq E$ and $N_{E'} \geq K$ where the coefficient of the balancing term $\lambda \geq 0$.

By optimizing the objective function, ERS clusters the pixels of homogeneous regions.

Easy samples refer to the points with the same prediction results from two networks, and these points usually have similar characteristics with the train set samples, so direct use of the predicted results of simple samples as pseudo labels has a high degree of confidence. The pseudo labels generated from the easy samples are used to improve the robustness of the classifier.

Difficult samples refer to the sample where GCN and deep neural network are inconsistent in prediction. We believe that excessive spectral variation leads to the generation of difficult samples. In order to reduce the probability of false label generation and improve reliability, we combined superpixel segmentation to generate pseudo labels for these samples: the image was divided into several subregions by superpixel segmentation. Samples clustered in the same superpixel often have great similarity, which determines the probability that they belong to the same category is high. Thus, for a hard sample, we used the deep neural network to predict all samples in the same region with it and then made majority voting (MV) to determine pseudo labels with high confidence for it. In our method, considering that superpixels contain robust spatial contextual information, high-confidence pseudo labels are generated for the selected hard samples by combining ERS and the MV strategy. The addition of these pseudo labels also helps the model to overcome the problems of different spectra and foreign bodies of the same spectrum. For example, let $S = [s_1, s_2, \ldots, s_K]$ represent a superpixel, which has K samples, and a hard sample s_e in S is labeled as follows:

$$\mathcal{L}\left(s_e\right) = \text{argmax} \sum_{y\in S_k} \delta(\mathcal{L}(y), c), \quad c = 1, 2, \ldots, C.$$

The purpose of adding difficult samples is to increase the generalization of the classifier.

C. 3D-Cross Attention Block

Inspired by the attention mechanism, we design a unique 3D-Cross Attention Block to achieve better joint feature expression of spatial information from hyperspectral and elevation information from LiDAR. Specifically, we consider that it is not only important in the spatial dimension but also the channel dimension to joint spatial features, so we use a 3-D attention mask to assign complementary attention. First, we concatenate all the spatial feature maps from HSI and LiDAR data to integrate all the spatial information, and then, the 3D-Cross Attention Block is responsible for enhancing the spatial features. Unlike other attention modules, 3D-Cross Attention Block extracts the attention mask by two parallel branches: 1-D convolution is used to extract the spatial attention mask from the channel dimension, 2-D convolution is used to extract the attention mask from the spatial dimension, and then, the attention masks of the two branches are allocated to the spatial features of the multisource data. It is worth mentioning that we assign these two masks synchronously, which can be viewed as 3-D mask elementwise multiply joint spatial feature maps. Denoted input features as F_{in} and output features as F_{out}, 3D-Cross Attention Block can be formulated as follows:

$$\begin{aligned} A_t &= \sigma\left(W_2 \delta\left(W_1 \,\text{GAP}(F_{\text{in}})\right)\right) \\ A_c &= \text{GAP}(\sigma\left(W_3\left(W_2 \delta\left(W_1 \,\text{GAP}(F_{\text{in}})\right)\right)\right)) \\ F_{\text{out}} &= F_{\text{in}} \otimes A_t \otimes A_c \end{aligned}$$

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where A_t and A_c denote the extracted spatial dimension attention features and channel dimension attention features, respectively, and $\otimes$ denotes elementwise multiply.

The 3D-Cross Attention Block guides the model to extract the regions of interest in the feature maps and makes the network pay attention to important features by assigning attention masks while suppressing useless information. Specifically, in the channel dimension, the input is compressed into 1×1 feature maps through 2-D global average pooling, and then, the attention mask is generated through three 1-D convolution/activation units. The output is regarded as a vector. In the spatial dimension, we directly use two 2-D convolution/activation units to extract the attention mask. The strategy of synchronous allocation generates a 3-D attention mask, which creates a cross relationship between the deep learning features that were originally fused only by concatenating and helps to make full use of the complementary features of multisource heterogeneous data in high-dimensional space.

IV. Experiment

In this section, we conducted experiments on the Houston 2013 dataset, Trento dataset, and MUUFL Guflport dataset. In order to quantitatively evaluate the classification performance of TSDN proposed in this article, we compare it with several hyperspectral classification algorithms and Hyperspectral–LiDAR fusion classification algorithms. All experimental results were obtained based on a server with an Intel Core I9-10900X CPU and four Nvidia RTX 2080Ti GPUs. Our deep learning framework versions used for the code are TensorFlow 1.14.0 and Keras 2.1.5. Overall accuracy (OA), average accuracy (AA), and Kappa coefficient are used as indicators.

This section first introduces the dataset used in the experiment and then compares the classification performance of eight algorithms mentioned in this article. Next, we do a detailed hyperparameter analysis and prove the effect of the 3D-Cross Attention module and LCM through an ablation study. Finally, we compare the complexity of several fusion classification algorithms.

A. Experiment Dataset

1) Houston2013 Dataset: The Houston2013 dataset was presented at the "IEEE GRSS Data Fusion Contest 2013." The scene was captured by the National Airborne Centre for laser mapping with its onboard ITRES CASI-1500 sensor at the University of Houston campus and neighboring areas in TX, USA. The dataset has a spatial dimension of 349×1905 and a spatial resolution of 2.5 m. HSI includes 144 spectral bands, covering a spectral wavelength range of 364–1046 nm. The LiDAR data contain one band. Groundtruth has labeled 15 classes, with a total of 15 029 samples, and the number of training sets is 2832 and the number of test sets is 12 197. The detailed categories and sample information are shown in Table I.

TABLE I
SAMPLE DISTRIBUTION OF HOUSTON2013 DATASET

No.	Color	Land-cover	Training	Test
1		Health grass	198	1053
2		Stressed grass	190	1064
3		Synthetic grass	192	505
4		Tress	188	1056
5		Soil	186	1056
6		Water	186	143
7		Residential	196	1072
8		Commercial	191	1053
9		Road	193	1059
10		Highway	191	1036
11		Railway	181	1054
12		Parking lot 1	192	1041
13		Parking lot 2	184	285
14		Tennis court	181	247
15		Running track	187	473
-	-	Total	2832	12197

TABLE II
SAMPLE DISTRIBUTION OF TRENTO DATASET

No.	Color	Land-cover	Training	Test
1		Apple trees	129	3905
2		Buildings	125	2778
3		Ground	105	374
4		Woods	154	8969
5		Vineyard	184	10317
6		Roads	122	3525
-	-	Total	819	29395

2) Trento Dataset: The Trento dataset was acquired using AISA Eagle sensors and shows a scene in the rural area of Trento, Italy. The spatial size of the dataset is 166×600, the spatial resolution is 1.0 m, and the spectral resolution is 9.2 nm. HSI includes 63 bands ranging from 420.89 to 989.09 nm. The LiDAR data contain one band. Groundtruth has labeled six classes, with a total of 30 214 samples; the number of training sets is 819 and the number of test sets is 29 535. The detailed categories and sample information are shown in Table II.

3) MUUFL Gulfport Dataset: The MUUFL Gulfport dataset [51], [52] was collected at 3500 ft and covered the campus of University of Southern Mississippi Gulf Park, Long Beach Mississippi, in November 2010. Due to noise, the dataset was clipped to a spatial size of 325×220 with a spatial resolution of 1 m. There are 72 bands in HSI, and noise bands 1–4 and 69–72 are generally removed in the experiment. The LiDAR data contain two bands. Groundtruth has labeled 11 classes, with a total of 53 687 samples. In our experiment, 100 points were randomly selected from each class as the training set and the remaining points as the test set. The calculation cost of adjacency matrix is too high due to the large number of labeled samples in this dataset, and we randomly select some points from the whole imagery to construct an adjacency matrix and carry out pseudo label generation in these samples. The detailed categories and sample information are shown in Table III.

B. Classification Performance

Four hyperspectral classification algorithms (labeled with H) and three fusion classification algorithms are selected as

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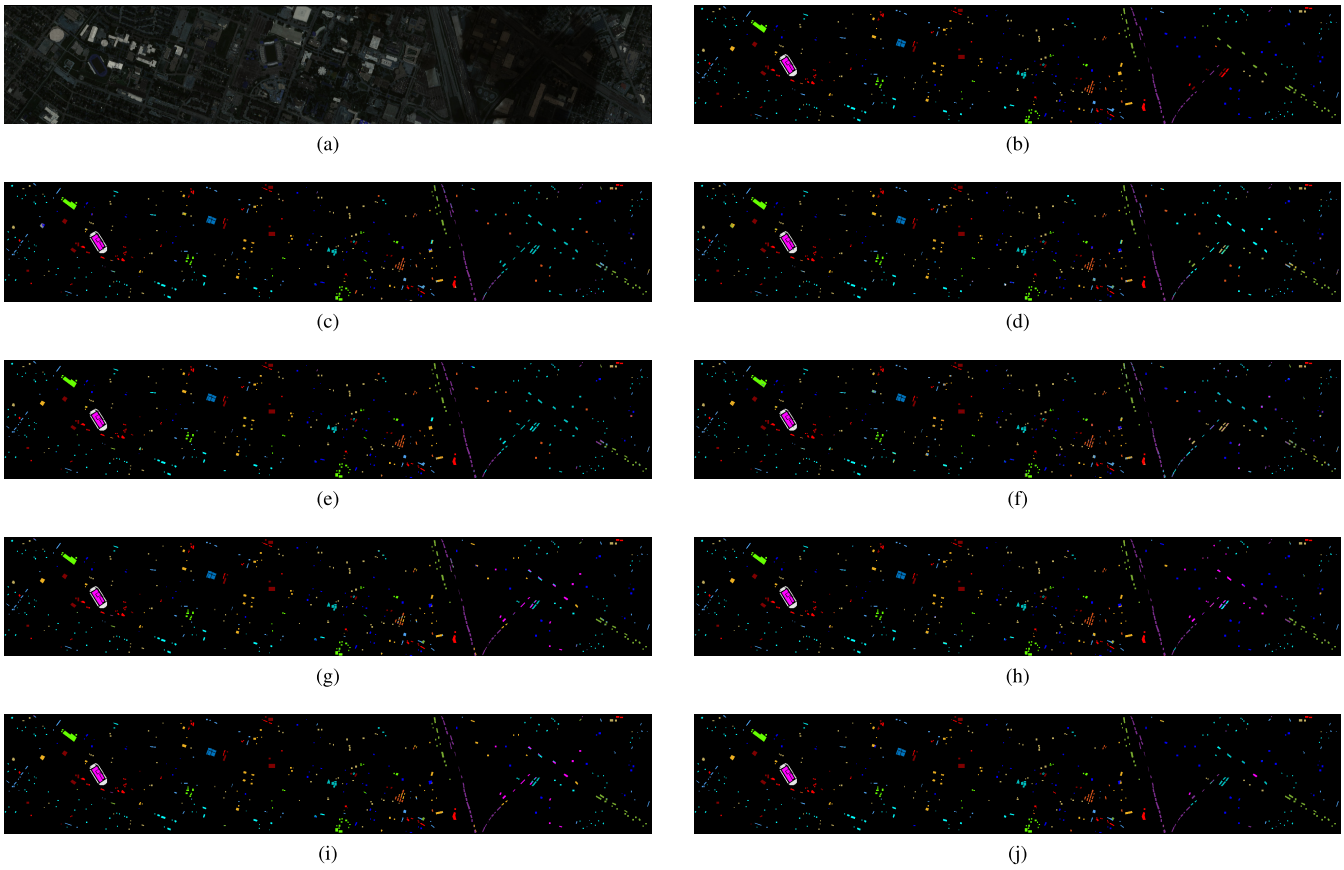


Fig. 5. Classification maps obtained via various algorithms for the Houston2013 dataset. (a) False-color image (bands 30, 45, and 65). (b) Groundtruth. (c) SVM. (d) ELM. (e) CNN-PPF. (f) GCN. (g) Two-Branch CNN. (h) FusAtNet. (i) HRWN. (j) TSDN.

TABLE III
SAMPLE DISTRIBUTION OF MUUFL GUFLPORT DATASET

No.	Color	Land-cover	Training	Test
1		Trees	100	23146
2		Grass Pure	100	4170
3		Grass Groundsurface	100	6782
4		Dirt and Sand	100	1726
5		Road Materials	100	6587
6		Water	100	366
7		Buildings' Shadow	100	2133
8		Buildings	100	6140
9		Sidewalk	100	1285
10		Yellow Curb	100	83
11		Cloth Panels	100	169
-	-	Total	1100	52587

comparative algorithms in our experiment. Among these comparison methods, the SVM and the ELM are traditional algorithms, the CNN-PPF is the hyperspectral classification algorithm based on deep learning, and the GCN represents semisupervised hyperspectral classification algorithms. The two-branch CNN, the FusAtNet, and the HRWN are high-precision fusion classification methods. On each dataset, all algorithms use the same training samples and test samples.

Tables IV, V, and VI quantitatively evaluate the classification results of the above algorithms on different datasets. Figs. 5, 6, and 7 show the classification map of each dataset in turn.

As traditional machine learning algorithms, the performance of the SVM and the ELM is similar and is significantly lower than deep learning-based algorithms. This is due to the poor extraction ability of traditional algorithms for data depth features. The CNN-PPF ignores the spatial context information of hyperspectral data. The training and deduction of the GCN depend on the adjacency matrix, in which the calculation cost increases sharply with the increase of samples. Overall, the classification results of fusion classification algorithms are generally better than those of normal hyperspectral classification algorithms because the introduction of elevation information has a great inhibitory effect on spectral aberration. The two-branch CNN algorithm uses two branches to extract deep learning features of heterogeneous data, which lays a foundation for subsequent methods. The FusAtNet leads to cross attention to consider the interaction of multisource data, thus improving the performance of the network. The HRWN algorithm achieves high-precision land-cover classification through pixel-level fusion and decision-level fusion. The TSDN algorithm in this article greatly expands the available train dataset and solves the problem of lacking labeled samples in the hyperspectral classification. The TSDN also has a very robust generalization capability, so it has superior classification performance on each dataset.

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TABLE IV
COMPARISON OF THE PERFORMANCE OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

class_id	class	SVM(H)	ELM(H)	CNN-PPF(H)	GCN(H)	Two-Branch	FusAtNet	HRWN	TSDN
1	Health grass	97.09%	97.33%	83.10%	82.72%	82.53%	**98.86%**	83.10%	83.10%
2	Stressed grass	98.07%	**99.32%**	84.02%	85.15%	85.06%	98.59%	85.15%	85.15%
3	Synthetic grass	**100.00%**	**100.00%**	**100.00%**	99.01%	**100.00%**	48.65%	99.41%	**100.00%**
4	Tress	97.58%	91.38%	92.80%	99.24%	93.75%	**100.00%**	99.72%	**100.00%**
5	Soil	97.00%	97.57%	99.91%	**100.00%**	99.43%	99.72%	**100.00%**	**100.00%**
6	Water	10.69%	**100.00%**	95.80%	**100.00%**	84.62%	80.12%	95.80%	98.60%
7	Residential	85.44%	54.35%	85.07%	86.66%	87.50%	84.18%	90.30%	**92.63%**
8	Commercial	66.51%	69.80%	44.82%	64.77%	78.06%	83.99%	**97.60%**	97.44%
9	Road	75.66%	76.38%	82.72%	64.88%	83.29%	**95.42%**	89.52%	94.05%
10	Highway	**85.53%**	63.95%	44.88%	76.25%	62.26%	85.12%	68.53%	68.53%
11	Railway	88.26%	81.12%	77.80%	78.75%	82.45%	76.74%	92.98%	**97.91%**
12	Parking lot 1	83.48%	78.51%	92.89%	50.43%	88.09%	84.73%	91.35%	**98.75%**
13	Parking lot 2	33.85%	38.97%	81.40%	37.54%	2.81%	**96.60%**	90.18%	91.23%
14	Tennis court	89.49%	88.77%	99.19%	**100.00%**	98.79%	**100.00%**	**100.00%**	**100.00%**
15	Running track	**100.00%**	96.69%	98.94%	**100.00%**	98.73%	96.84%	**100.00%**	**100.00%**
	OA	78.42%	79.83%	81.19%	80.28%	83.89%	87.10%	90.88%	**92.68%**
	AA	80.58%	82.28%	84.22%	81.68%	81.82%	88.64%	92.21%	**93.83%**
	Kappa	0.7678	0.7814	0.7975	0.7862	0.8253	0.8602	0.9041	**0.9207**

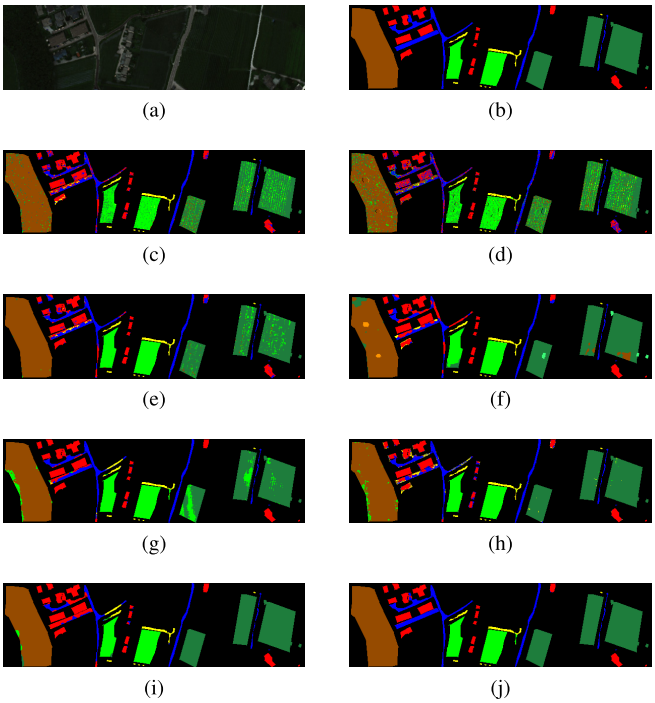


Fig. 6. Classification maps obtained via various algorithms for the Trento dataset. (a) False-color image (bands 12, 20, and 30). (b) Groundtruth. (c) SVM. (d) ELM. (e) GCN. (f) CNN-PPF. (g) Two-Branch CNN. (h) FusAtNet. (i) HRWN. (j) TSDN.

According to the above classification maps, the SVM and the ELM can describe the contour of the land-cover map, but there are also many internal noise points. Algorithms based on deep learning have less noise, but the area of misjudgment is still obvious. Generally speaking, the classification results of algorithms considering spatial context are continuous, so once misjudgment occurs, the area of noise will be larger than other algorithms. On the premise of ensuring the spatial continuity of the prediction results, our TSDN algorithm improves the classification accuracy and achieves a good visual effect.

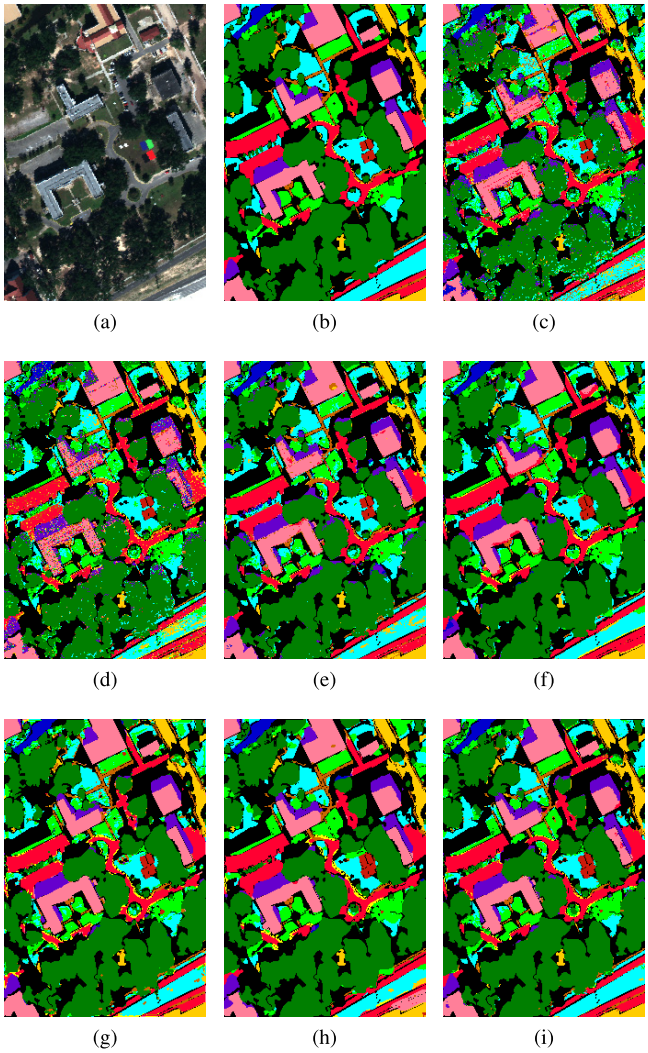


Fig. 7. Classification maps obtained via various algorithms for the MUUFL Gulfport dataset. (a) False-color image. (b) Groundtruth. (c) SVM. (d) ELM. (e) CNN-PPF. (f) Two-Branch CNN. (g) FusAtNet. (h) HRWN. (i) TSDN.

For the Houston2013 dataset, serious spectral distortion exists due to large scene coverage, sparse labeled samples,

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TABLE V
COMPARISON OF THE PERFORMANCE OF DIFFERENT METHODS ON THE TRENTO DATASET

class_id	class	SVM(H)	ELM(H)	CNN-PPF(H)	GCN(H)	Two-Branch	FusAtNet	HRWN	TSDN
1	Apple trees	64.14%	57.12%	99.64%	94.37%	97.87%	98.02%	99.31%	**99.95%**
2	Buildings	74.58%	75.38%	86.25%	89.63%	99.35%	88.09%	98.67%	**99.89%**
3	Ground	67.17%	68.71%	97.33%	**100.00%**	91.23%	94.86%	89.04%	93.32%
4	Woods	93.39%	81.98%	99.69%	95.36%	92.47%	96.84%	98.08%	**100.00%**
5	Vineyard	89.63%	82.37%	90.28%	96.48%	96.62%	98.88%	**100.00%**	**100.00%**
6	Roads	85.56%	76.55%	77.00%	63.17%	76.09%	88.95%	61.86%	**93.81%**
	OA	83.78%	76.05%	92.72%	91.80%	93.55%	96.10%	95.10%	**99.25%**
	AA	79.08%	73.69%	91.70%	89.84%	92.27%	94.27%	91.16%	**97.83%**
	Kappa	0.7857	0.6826	0.9033	0.8900	0.9147	0.9477	0.9344	**0.9900**

TABLE VI
COMPARISON OF THE PERFORMANCE OF DIFFERENT METHODS ON THE MUUFL GUFLPORT DATASET

class_id	class	SVM(H)	ELM(H)	CNN-PPF(H)	Two-Branch	FusAtNet	HRWN	TSDN
1	Trees	96.01%	95.12%	90.44%	93.07%	**96.80%**	94.66%	95.21%
2	Grass Pure	58.58%	56.48%	78.35%	**92.01%**	71.25%	88.47%	83.21%
3	Grass Groundsurface	71.81%	74.15%	80.92%	72.56%	**85.91%**	67.99%	76.87%
4	Dirt and Sand	65.88%	58.86%	93.97%	90.27%	88.34%	**96.47%**	94.50%
5	Road Materials	90.22%	86.31%	91.07%	93.29%	**98.25%**	86.59%	93.52%
6	Water	40.70%	61.93%	**100.00%**	95.90%	73.00%	**100.00%**	99.73%
7	Buildings' Shadow	46.96%	54.23%	86.45%	86.40%	86.73%	94.94%	**96.02%**
8	Buildings	89.07%	88.69%	89.38%	84.50%	**96.11%**	92.90%	94.32%
9	Sidewalk	39.57%	44.86%	71.75%	55.72%	50.50%	66.46%	**82.33%**
10	Yellow Curb	25.68%	16.70%	77.11%	64.47%	10.28%	84.34%	**92.77%**
11	Cloth Panels	70.76%	47.97%	97.63%	95.86%	60.95%	97.63%	**98.22%**
	OA	79.53%	79.96%	87.77%	88.08%	89.23%	88.93%	**91.31%**
	AA	63.20%	62.30%	87.01%	84.28%	74.37%	88.22%	**91.52%**
	Kappa	0.7953	0.7389	0.8404	0.8433	0.8578	0.8538	**0.8853**

TABLE VII
ABLATION STUDY ON THE HOUSTON2013 DATASET

	with 3D-Cross Attention Block	with LCM	OA
Base	×	×	86.93%
Base	✓	×	89.15%
Base	×	✓	90.54%
Base	✓	✓	**92.68%**

TABLE VIII
ABLATION STUDY OF ATTENTION BLOCKS

	OA	AA	Kappa
without any Attention	88.80%	90.97%	0.8790
without spectral Attention	91.72%	93.15%	0.9104
without 3D-Cross Attention	90.54%	92.19%	0.8977
TSDN	**92.68%**	**93.83%**	**0.9207**

TABLE IX
COMPARISON OF TRAINING TIME AND MODEL COMPLEXITY ON THE MUUFL GULFPORT DATASET WITH A 2080TI

	Two-Branch CNN	FusAtNet	HRWN	TSDN
model/kb	37602	439253	14237	20862
train/s	24.9	8295.39	24.49	27.06
test/s	27.82	36.41	19.38	33.45

and multiple sample categories. In this method, the model generalization ability is improved and the classification accuracy is improved by introducing difficult samples through semisupervised learning.

For the Trento and MUUFL Gulfport datasets, the labels of dataset samples are dense, and the model prediction is prone to noise. This method can improve the continuity of spatial information and solve this problem well by introducing easy samples through semisupervised learning. However, for GCN, the increase of labeled samples increases the computation cost index level of adjacency matrix. Therefore, only a part of samples are randomly selected for GCN training in the pseudo labels generation process of MUUFL Gulfport dataset.

Based on all the above experimental results, TSDN has good generalization to two different types of dataset with dense labeled and sparse labeled, thus achieving good classification performance. From the classification map, TSDN prediction results have excellent spatial continuity, which is almost no noise. On the one hand, the 3D-Cross Attention Block enhances the network's ability to perceive spatial information; on the other hand, the pseudo labels generated by LCM are randomly distributed on the whole dataset to guide the model to discriminate unknown areas and reduce the discipline of noise.

C. Parameter Analysis

Fig. 8 shows the joint influence of pseudo sample rate and training set percent on accuracy. Different hyperparameter pairs have slightly different effects on different datasets. In Tables IV, V, and VII, we set pseudo samples rate as 1.0 and training set percent as 100%. It is worth mentioning

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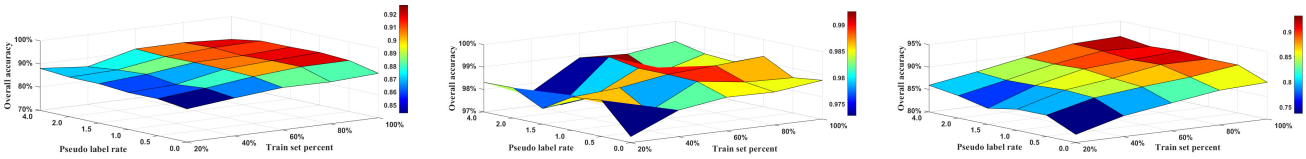


Fig. 8. Performance of different pseudo sample rates and train set percents. Rate means the number of pseudo samples divided by the number of real samples.

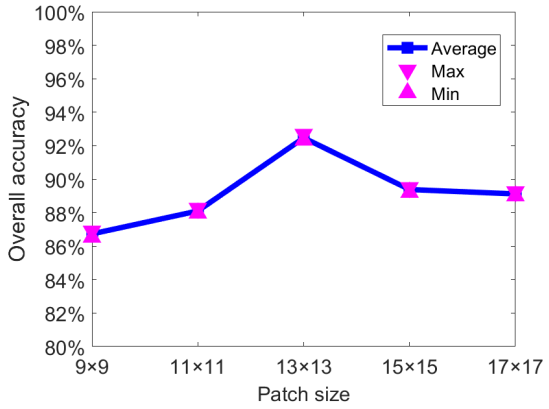


Fig. 9. Performance of different spatial patch sizes on Houston2013.

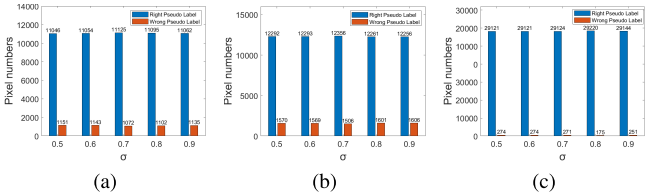


Fig. 10. Effects of superpixel segmentation parameters on pseudo sample accuracy. (a) Houston2013. (b) Trento. (c) MUUFL Guflport.

that experiments show that our proposed TSDN is relatively insensitive to the number of training samples, which means that our model has better classification performance while lacking training samples. In general, on the one hand, with the increase of real samples, the accuracy of the network becomes higher and higher because the increase of samples will make the classifier have more accurate clustering centers. On the other hand, when the pseudo label rate is low, the expansion of the train set is not obvious, the generalization of the network is low, and therefore, the classification accuracy is low. Also, when the rate is high, the accumulation of error samples in pseudo samples misleads the direction of network convergence, so the classification accuracy will decrease.

Fig. 9 shows the influence of different spatial patch sizes. The experiment on Houston2013 shows that when the patch size is 13 × 13, TSDN obtains the highest OA.

Fig. 10 shows the influence of different superpixel segmentation parameters on the LCM performance. In the above experiments, σ was fixed at 0.7.

D. Ablation Study

Table VI shows the influence of 3D-Cross Attention Block and LCM module proposed in this article on classification over accuracy. The first line shows the classification performance of the basic model, the second and third lines show the improvement of OA by 3D-Cross Attention Block (about 3%) and LCM (about 4%), respectively, and the fourth line shows that when 3D-Cross Attention Block and LCM are used at the same time, the classification result is the best (improve by about 6%). In fact, the 3D-Cross Attention Block enhances the complementarity of multisource information in the feature-level fusion process and improves the model's ability to represent deep features. The LCM not only enables the model to identify samples with large spectral distortion but also significantly reduces the noise in the classification result graph and optimizes the prediction boundary of the model by introducing difficult and easy samples with high confidence.

Since our proposed TSDN not only uses the 3D-Cross Attention Block to enhance joint spatial features but also uses the spectral attention mechanism to enhance spectral features, we combine deep learning features in different ways by canceling each attention module to further verify the role of these modules. Table VIII shows the classification performances of the corresponding ablation experiments. Based on these experimental results, we believe that all the attention modules in the proposed TSDN are necessary for accuracy improvement.

Therefore, the ablation study described above met our expectations in designing these modules.

E. Computational Complexity

Table IX reports the evaluation of timings of four fusion classification algorithms on the MUUFL Guflport dataset. Although our TSDN method introduces some additional running complexity and a slightly larger model size compared with other methods, it achieves much better accuracy. On the whole, the calculation cost required by TSDN is limited to an acceptable range.

V. Conclusion

In this study, in order to fuse hyperspectral and LiDAR more effectively, we propose a triplet branch semisupervised deep network, which uses three independent CNN branches to extract deep features. To solve the problem of lack of samples, the LCM module proposed in this article adopts the results of the pretrain model and GCN to achieve the division of hard and easy samples, then guides the generation of pseudo labels based on the ERS superpixel segmentation theory, and finally

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realizes semisupervised training. This article also proposes a 3D-Cross Attention Block to enhance the complementarity of multisource information. Results on three public datasets show that this method has superior classification performance. It is worth noting that the idea of this study can also be extended to semantic segmentation, target detection, or other fields.

REFERENCES

[1] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[2] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[3] N. Yokoya, J. Chan, and K. Segl, "Potential of resolution-enhanced hyperspectral data for mineral mapping using simulated EnMAP and Sentinel-2 images," *Remote Sens.*, vol. 8, no. 3, p. 172, Feb. 2016. [Online]. Available: https://www.mdpi.com/2072-4292/8/3/172

[4] S. Zhang, J. Li, Z. Wu, and A. Plaza, "Spatial discontinuity-weighted sparse unmixing of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5767–5779, Oct. 2018.

[5] W. Liao, A. Pizurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[6] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[7] W. Li, C. Chen, H. Su, and Q. Du, "Local binary patterns and extreme learning machine for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3681–3693, Jul. 2015.

[8] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[9] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[10] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[11] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[12] B. Xi, J. Li, Y. Li, R. Song, W. Sun, and Q. Du, "Multiscale context-aware ensemble deep KELM for efficient hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5114–5130, Jun. 2021.

[13] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[14] D. Wang, B. Du, L. Zhang, and Y. Xu, "Adaptive spectral-spatial multiscale contextual feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2461–2477, Mar. 2020.

[15] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[16] B. Xi et al., "Semisupervised cross-scale graph prototypical network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 23, 2022, doi: 10.1109/TNNLS.2022.3158280.

[17] Y. Li, B. Xi, J. Li, R. Song, Y. Xiao, and J. Chanussot, "SGML: A symmetric graph metric learning framework for efficient hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 609–622, 2022.

[18] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[19] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[20] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[21] J. Jung, E. Pasolli, S. Prasad, J. C. Tilton, and M. M. Crawford, "A framework for land cover classification using discrete return LiDAR data: Adopting pseudo-waveform and hierarchical segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 2, pp. 491–502, Feb. 2014.

[22] M. Zhang, P. Ghamisi, and W. Li, "Classification of hyperspectral and LiDAR data using extinction profiles with feature fusion," *Remote Sens. Lett.*, vol. 8, no. 10, pp. 957–966, Oct. 2017.

[23] M. Belgiu, I. Tomljenovic, T. Lampoltshammer, T. Blaschke, and B. Höfle, "Ontology-based classification of building types detected from airborne laser scanning data," *Remote Sens.*, vol. 6, no. 2, pp. 1347–1366, Feb. 2014.

[24] I. Tomljenovic, B. Höfle, D. Tiede, and T. Blaschke, "Building extraction from airborne laser scanning data: An analysis of the state of the art," *Remote Sens.*, vol. 7, no. 4, pp. 3826–3862, Mar. 2015.

[25] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[26] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[27] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[28] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[29] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based SpectroSpatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 416–425.

[30] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[31] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[32] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[33] H.-C. Li, W.-S. Hu, W. Li, J. Li, Q. Du, and A. Plaza, "A^3CLNN: Spatial, spectral and multiscale attention ConvLSTM neural network for multisource remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 747–761, Feb. 2022.

[34] Z. Feng, Q. Zhou, G. Cheng, X. Tan, J. Shi, and L. Ma, "Semi-supervised semantic segmentation via dynamic self-training and class-balanced curriculum," 2020, *arXiv:2004.08514*.

[35] R. Mendel, L. A. D. Souza, D. Rauber, J. P. Papa, and C. Palm, "Semi-supervised segmentation based on error-correcting supervision," in *Proc. ECCV*, 2020, pp. 141–157.

[36] M. S. Ibrahim, A. Vahdat, M. Ranjbar, and W. G. Macready, "Semi-supervised semantic image segmentation with self-correcting networks," 2018, *arXiv:1811.07073*.

[37] Y. Zhu et al., "Improving semantic segmentation via self-training," 2020, *arXiv:2004.14960*.

[38] B. Zoph et al., "Rethinking pre-training and self-training," 2020, *arXiv:2006.06882*.

[39] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[40] C. Persello and L. Bruzzone, "Active and semisupervised learning for the classification of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6937–6956, Nov. 2014.

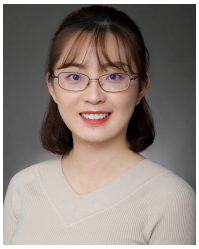
[41] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[42] D. Liao, Y. Qian, J. Zhou, and Y. Y. Tang, "A manifold alignment approach for hyperspectral image visualization with natural color," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3151–3162, Jun. 2016.

[43] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," 2014, *arXiv:1409.0473*.

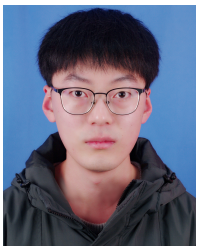
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[44] A. Ambartsoumian and F. Popowich, "Self-attention: A better building block for sentiment analysis neural network classifiers," 2018, *arXiv:1812.07860*.
[45] H. Jie, S. Li, S. Gang, and S. Albanie, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2017.
[46] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.
[47] S. Subudhi, R. N. Patro, P. K. Biswal, and F. Dell'Acqua, "A survey on superpixel segmentation as a preprocessing step in hyperspectral image analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5015–5035, 2021.
[48] T. Priya, S. Prasad, and H. Wu, "Superpixels for spatially reinforced Bayesian classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 5, pp. 1071–1075, May 2015.
[49] Y. Zhang, K. Liu, Y. Dong, K. Wu, and X. Hu, "Semisupervised classification based on SLIC segmentation for hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1440–1444, Aug. 2020.
[50] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. CVPR*, Jun. 2011, pp. 2097–2104.
[51] X. Du and A. Zare, "Technical report: Scene label ground truth map for MUUFL Gulfport data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep., 20170417, 2017.
[52] P. G. M. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL Gulfport hyperspectral and LiDAR airborne data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep., REP-2013-570, 2013.



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